# Transformations, congruence, and similarity

## Core Concepts: Transformations, congruence, and similarity

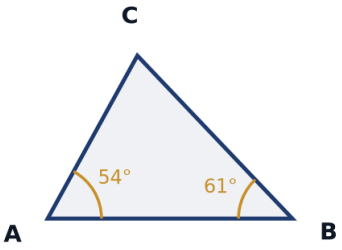
**1** In $\triangle ABC$ studying transformations, congruence, and similarity, $m\angle A = 51^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

**a.** $84^\circ$

**b.** $94^\circ$

**c.** $76^\circ$

**d.** $129^\circ$




**2** A geometric figure scaled by factor $k = 2$ has an original side length of $5$ cm. What is the corresponding image length?

**a.** 8 cm

**b.** 10 cm

**c.** 15 cm

**d.** 20 cm

**3** On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its $y$-coordinate when $x = 1$?

**a.** 3

**b.** 8

**c.** 6

**d.** 9

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## Procedural Fluency & Algebraic Manipulation

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**4** In $\triangle ABC$ studying transformations, congruence, and similarity, $m\angle A = 51^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

**a.** $77^\circ$

**b.** $87^\circ$

**c.** $69^\circ$

**d.** $129^\circ$

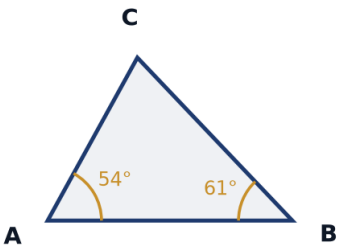
**5** A geometric figure scaled by factor $k = 3$ has an original side length of $5$ cm. What is the corresponding image length?

**a.** 12 cm

**b.** 15 cm

**c.** 20 cm

**d.** 45 cm




**6** On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its $y$-coordinate when $x = 2$?

**a.** 6

**b.** 11

**c.** 9

**d.** 12

## Graphical & Geometric Interpretation

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**7** In $\triangle ABC$ studying transformations, congruence, and similarity, $m\angle A = 51^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

**a.** $70^\circ$

**b.** $80^\circ$

**c.** $62^\circ$

**d.** $129^\circ$

**8** A geometric figure scaled by factor $k = 4$ has an original side length of $5$ cm. What is the corresponding image length?

**a.** 16 cm

**b.** 20 cm

**c.** 25 cm

**d.** 80 cm

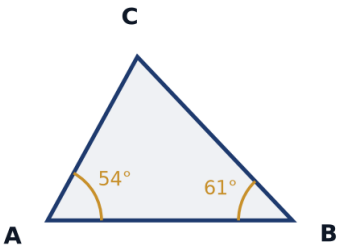
**9** On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its $y$-coordinate when $x = 3$?

**a.** 9

**b.** 14

**c.** 12

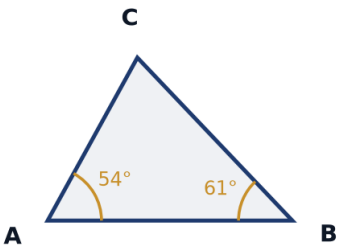
**d.** 15




## Exam-Style Problem Solving & Modeling

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**10** In $\triangle ABC$ studying transformations, congruence, and similarity, $m\angle A = 51^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?




**a.** $63^\circ$

**b.** $73^\circ$

**c.** $55^\circ$

**d.** $129^\circ$

**11** A geometric figure scaled by factor $k = 5$ has an original side length of $5$ cm. What is the corresponding image length?

**a.** 20 cm

**b.** 25 cm

**c.** 30 cm

**d.** 125 cm

**12** On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its $y$-coordinate when $x = 4$?

**a.** 12

**b.** 17

**c.** 15

**d.** 18

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